

Engineering Electromagnetics II

Homework #2

90 points total

ECE 332 – Spring 2026

Due Tuesday, May 5, 2026

1. Write general expressions for the electric and magnetic fields of a 1 GHz sinusoidal plane wave traveling in the $+y$ direction in a lossless non-magnetic medium with relative permittivity $\epsilon_r = 9$. The electric field is polarized along the x direction, its peak value is 6 V/m, and its intensity is 4 V/m at $t = 0$ and $y = 2$ cm. (10 points)
2. The electric field of a plane wave propagating in a lossless, nonmagnetic, dielectric material with $\epsilon_r = 2.56$ is given by

$$\mathbf{E} = 20 \cos \left((6\pi \times 10^9) t - kz \right) \hat{\mathbf{y}} \text{ (V/m)}.$$

(12 points total)

Determine

- (a) f , u_p , λ , k , and η . (5 points)
 - (b) the magnetic field $\mathbf{H}$. (7 points)
3. For a wave characterized by the electric field

$$\mathbf{E}(z, t) = a_x \cos(\omega t - kz) \hat{\mathbf{x}} + a_y \cos(\omega t - kz + \delta) \hat{\mathbf{y}},$$

identify the polarization state, determine the polarization angles (γ, χ) , and sketch the locus of $\mathbf{E}(0, t)$ for each of the following cases.

(16 points total)

- (a) $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = 0$ radians (4 points)
- (b) $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = \pi$ radians (4 points)
- (c) $a_x = 3$ V/m, $a_y = 3$ V/m, and $\delta = \pi/4$ radians (4 points)
- (d) $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = -3\pi/4$ radians (4 points)

4. A linearly polarized plane wave of the form

$$\tilde{\mathbf{E}} = a_x e^{-jkz} \hat{\mathbf{x}}$$

can be expressed as the sum of an RHC-polarized wave with magnitude a_R and an LHC-polarized wave with magnitude a_L . Prove this statement by finding expressions for a_R and a_L in terms of a_x . (8 points)

5. Dry soil is characterized by $\epsilon_r = 2.5$, $\mu_r = 1$, and $\sigma = 10^{-4}$ S/m. At each of the following frequencies, determine if dry soil may be considered a good conductor, a quasi-conductor, or a low-loss dielectric. Then calculate α , β , λ , u_p , and η_c . (24 points total)

- (a) 60 Hz (6 points)
- (b) 1 kHz (6 points)
- (c) 1 MHz (6 points)
- (d) 1 GHz (6 points)

6. In a nonmagnetic, lossy, dielectric medium, a 300 MHz plane wave is characterized by the magnetic field phasor

$$\tilde{\mathbf{H}} = (\hat{\mathbf{x}} - j4\hat{\mathbf{z}})e^{-2y}e^{-j9y} \quad (\text{A/m})$$

Obtain time-domain expressions for the electric and magnetic field vectors. (10 points)

7. A wave traveling in a nonmagnetic medium with $\epsilon_r = 9$ is characterized by an electric field given by

$$\mathbf{E} = 3 \cos \left((\pi \times 10^7) t + kx \right) \hat{\mathbf{y}} - 2 \cos \left((\pi \times 10^7) t + kx \right) \hat{\mathbf{z}} \quad (\text{V/m}).$$

Determine the direction of wave travel and average power density carried by the wave. (10 points)